

CHAPTER – 06

TUPLES

What is a Tuple?

- A **tuple** is an **ordered** collection of elements stored in a single variable.
- It is **immutable**, meaning elements **cannot be changed** after the tuple is created.
- Tuples allow **duplicate elements**.
- Defined using **round brackets ()**.

2. Tuple Characteristics

Feature	Description
Ordered	Maintains element order
Immutable	Cannot change or modify elements
Can contain duplicates	Duplicate values allowed
Mixed data types	Can contain elements of different types

3. Creating Tuples

- **Multiple elements:**

```
t = ("apple", "banana", "kiwi")
```

- **Single element tuple:**

Use comma , after the element to distinguish from string:

```
t1 = ("apple",) # Tuple with one element
```

```
t2 = ("apple") # String, not tuple
```

4. Indexing & Accessing Tuple Elements

- Zero-based indexing: first element at index 0.
- Access by index:

```
t = ("a", "b", "c")
```

```
print(t[1]) # Output: b
```

5. Immutability & Modifying Tuples (Workaround)

- Tuples **cannot be modified** directly.
- To "modify," convert to a list, make changes, convert back to tuple:

```
CODE : t = ("apple", 2, 5, "banana")
```

```
temp_list = list(t)
```

```
temp_list[1] = 10.5
```

```
t = tuple(temp_list)
```

6. Adding Items to Tuples

- Tuples themselves can't be changed, but **concatenation creates new tuples**:

```
CODE: t1 = ("apple", "kiwi")
```

```
t2 = ("orange",)
```

```
t3 = t1 + t2 # New tuple formed
```

7. Packing and Unpacking Tuples

- **Packing:** Assign multiple values to one tuple variable.

```
items = (1, 2, 3)
```

- **Unpacking:** Assign tuple elements to multiple variables.

```
a, b, c = items
```

- Number of variables must match number of elements.

8. Unpacking with Asterisk *

- When variable count differs, use * to capture remaining elements as a list.

```
items = (1, 2, 3, 4, 5)
```

```
a, b, *c = items
```

```
# a=1, b=2, c=[3, 4, 5]
```

9. in Keyword with Tuples

- Check if an element exists inside a tuple.

```
t = ("a", "b", 1, 2)
```

```
if 1 in t:
```

```
    print("Present")
```

else:

```
print("Not present")
```

10. Common Tuple Methods & Functions

Method/Function	Description
len(t)	Returns length of tuple
t.count(x)	Counts occurrences of x
t.index(x)	Returns index of first occurrence of x

Practice Set – Tuples (Only Questions)

1. Create a tuple with mixed data types (string, integer, float). Access and print the element at the second index.
2. How do you create a tuple with only one element? Write an example.
3. Given a tuple, write steps to change its second element from "banana" to "orange" using a workaround.
4. Write code to combine two tuples ("apple", "banana") and ("kiwi", "grapes") into a single tuple.
5. Explain what will happen if you try to modify an element of a tuple directly, with an example.
6. Using tuple unpacking, assign the values of the tuple (10, 20, 30) to three variables a, b, and c.
7. How does the asterisk (*) work in tuple unpacking? Give an example.
8. Write a Python expression using the in keyword to check whether the element "apple" exists in the tuple ("banana", "orange", "apple").
9. Given the tuple t = (1, 2, 3, 4, 5), use unpacking with * to assign the first two elements to variables a and b, and the remaining elements to a list c.
10. After converting a tuple to a list, which list methods can be used without causing errors? Which methods would not work on the tuple itself? Explain briefly.